

Cross-Track Collaboration

Track I Collaborated With: Data Science Track

What Information Was Exchanged:

What I Shared (Data Analytics)

My final dataset (after cleaning missing values)

My KPI calculations and findings

Which variables seemed most important (lead time, distance, reminders)

My insights about which patient groups have highest no-show rates

Data quality issues I found

What They Shared (Data Science)

Their target variable definition (No-Show)

Their proposed features for the prediction model

Their modelling approach (classification model)

How they plan to handle cancellations in their model

Their train/test split strategy

What I Shared (Data Analytics)

My dashboard visualizations

What They Shared (Data Science)

Which metrics they'll use to evaluate the model

Why This Collaboration Was Relevant:

The Data Science track is building a **predictive model** to identify patients at high risk of no-show. My analysis provides the foundation for their work because:

- **I identified which variables matter most** (lead time, distance, reminders) → They know what features to include
- **I cleaned the data** → They don't have to start from scratch
- **I found patterns** → They can validate their model against my findings
- **My KPI definitions** → They use the same definitions for their target variable

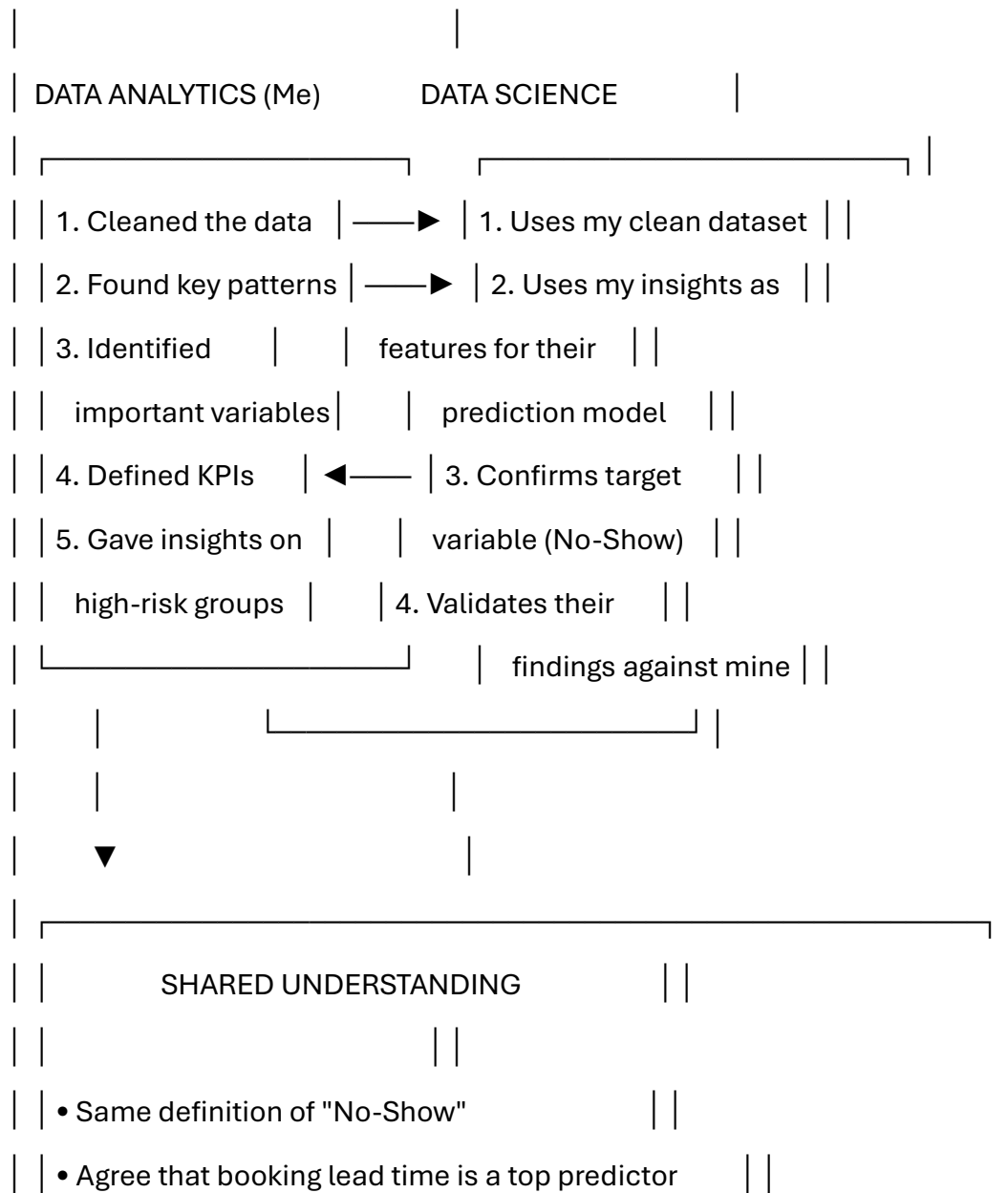
Simple Example:

I found that patients booking 1-2 months in advance have only 34% attendance. The Data Science team can now use "booking_lead_days" as a feature in their model to predict which patients will no-show.

How Our Work Connects:

text

OUR COLLABORATION FLOW



- Agree that distance is a key factor
- Reminders (especially WhatsApp) improve attendance
- Specialist consultations need extra attention

What Changed or Improved as a Result:

Before Collaboration

After Collaboration

I was analyzing data in isolation

I now know exactly what the Data Science team needs from my work

I didn't know how they would use my findings

I focused on variables that would be useful for prediction

My recommendations were general

My insights now feed directly into the predictive model

We had different definitions of "No-Show"

We agreed on one definition so our work is consistent

Specific Examples of Improvement:

1. Focused My Analysis:

Instead of analyzing everything, I gave extra attention to variables that the Data Science team will use as features in their model (lead time, distance, reminder_channel, previous_no_shows).

2. Cleaned Data Purposefully:

I handled missing values in a way that works for their model. For example, leaving "reminder_channel" blanks as missing (meaning "No Reminder Sent") is useful for prediction.

3. Validated Their Approach:

When they shared their modelling plan, I could confirm whether it made sense based on what I saw in the data. For example, I confirmed that "booking_lead_days" is indeed a strong predictor.

4. Avoided Duplication:

Instead of both of us doing the same work, I focused on understanding **what** is happening, and they focus on **predicting** what will happen.

Practical Example of Our Collaboration:

My Finding

Patients booking 1-2 months ahead have 34% attendance

Very Far Distance patients have 68% no-show rate

How Data Science Uses It

They use "booking_lead_days" as a feature in their model

They use "distance_to_clinic_km" to predict no-shows

My Finding

How Data Science Uses It

WhatsApp reminders give best attendance

They can test if "reminder_channel" improves model accuracy

Patients with previous no-shows are more likely to no-show again

They include "previous_no_shows" as a feature

Younger patients (18-24) have highest no-show rates

They can use "age_group" in their model

What I Learned from This Collaboration:

1. **My work has real value** beyond just reporting - it builds the foundation for predictive modelling
2. **I need to think about what will be useful** for other tracks, not just what's interesting to me
3. **Consistency matters** - we need to agree on definitions (like what counts as a "No-Show")
4. **Data cleaning is critical** - the Data Science team can't build a good model on bad data

Summary of Our Collaboration Value:

What I Gave to Data Science

Clean, ready to use dataset

List of important variables

Clear definitions of no-show

Insights on high-risk groups

Data quality documentation

What I Received from Data Science

Clear understanding of what they need from me

Validation that my insights are useful

Better focus in my analysis

Their modelling approach (so I can align my work)

Confirmation that my data prep was correct

Next Steps Together:

1. **Week 5:** I provide final insights and dataset. Data Science starts building baseline model.
2. **Week 6:** Data Science shares their model results. I help interpret the findings and suggest business actions.

This collaboration ensures that the work I did in Week 5 doesn't just sit in a report - it directly helps build a predictive tool that HealthConnect can use to reduce no-shows and improve patient care.